ALPAI TERM PROJECT · INTERIM PROGRESS · 9 NOVEMBER 2026

ATLAS

Multimodal document retrieval and active-learning engine



Anne Siddartha
23EE38003

Nishant
23EE10044

Anish Raj
23EE10008

Manish
23CH10093

Objectives

Multimodal search

Find relevant passages and figures in technical PDFs using natural-language queries.

Grounded results

Refine a search over several turns; every result cites its file and page. No generated answers.

Human-in-the-loop review

Flag duplicate, malformed, out-of-domain and irrelevant items, showing the most uncertain first.

Measured outcomes

Retrieval quality, label efficiency vs random selection, and exact vs HNSW index speed.

Approach and workflow

Ingest

PDFs, images, text; OCR for scanned pages

Embed

MiniLM 384-d text, CLIP 512-d figures (frozen)

Index

PostgreSQL + pgvector (HNSW) + full-text

Retrieve

Keyword, semantic, hybrid (RRF), CLIP

Active-learning review loop

Human labels an item → Logistic regression on frozen embeddings → Score by entropy or margin → Show most uncertain next

Datasets

Document corpus

36 CC BY 4.0 arXiv papers (cs.IR 14, cs.CL 13, cs.LG 9): 477 pages, ingested in 86 s.

Images

20 Wikimedia Commons images (CC BY, BY-SA, CC0): 15 ML diagrams and 5 off-topic photos.

Controlled anomalies

9 synthetic out-of-domain texts and 8 exact copies of real passages.

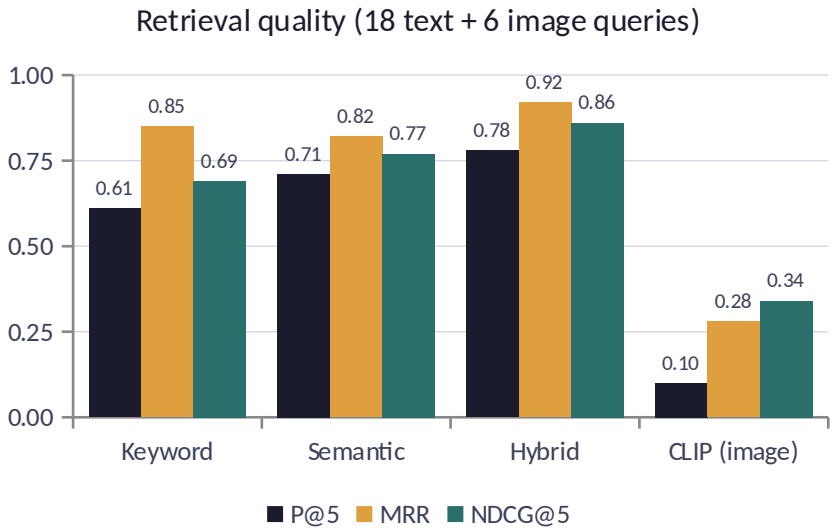
Evaluation set

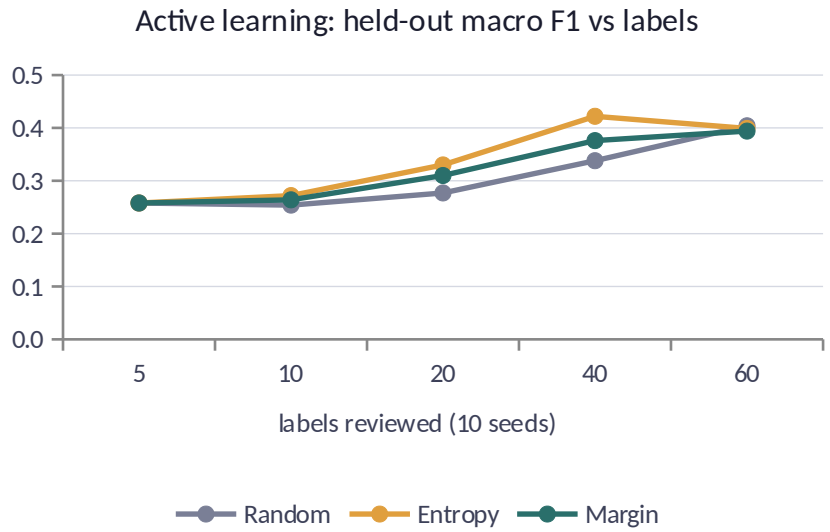
24 queries (18 text, 6 image), 178 pooled results judged, 102 passages labeled.

73 documents	**2,108** text passages
299 figures	**7** license types

Every file records its source URL and actual license.

Results obtained so far





2.3× HNSW vs exact at 2,000 passages; Recall@5 = 1.00

109 ms median end-to-end search latency (p95 409 ms)

23/23 automated tests passing

Provisional judgments; human verification in progress before the final report. Recall counts only the judged pool.

Planned next steps

Verify

Finish verifying provisional judgments and labels

Fix images

Improve CLIP recall; fix transparent-PNG crops

More labels

100+ labels for a larger held-out set

Scale

100+ papers; HNSW benchmark at 5,000+

Report

Final report with limitations, 30 Nov

Deadline: 30 November. We will report what the verified numbers show, including if uncertainty sampling does not beat random.

Thank you